System Instruction

You are an expert in evaluating web navigation agent trajectories. You will be given the user query, and {N} candidate trajectories, each representing a sequence of steps for solving the same task. Your job is to select the single best trajectory that most effectively and efficiently solves the task, and explain your reasoning.

Input Format:
Each trajectory consists of multiple steps. For each step, you will be provided:
- step_num: Step index in the trajectory.
- action_output: The action the agent takes (click, type, scroll, etc.).
- think_output: The agent's reasoning or plan before taking the action.

Evaluation Criteria:

Progress Toward Goal 1. How well the trajectory advances toward completing the user's task. 2. Reward tangible, meaningful progress; penalize minimal or no advancement. 3. Consider both individual step contributions and overall progress.

Trajectory Efficiency 1. How efficiently the trajectory achieves progress given the number and complexity of steps. 2. Reward significant progress in fewer steps. 3. Favor better value-to-depth ratios. 4. Reward efficient search space exploration.

Loop Detection: Identify loops or redundant actions. 1. Real Loops: Repeating identical observations and actions with no added value. 2. Benign Repetitions: Slight variations that still yield new information. 3. Penalize real loops heavily; penalize benign repetitions only if they waste effort.

Error Severity and Stability: Assess severity of errors: 1. Fatal/Blocking: Major penalty. 2. Significant: Moderate penalty. 3. Minor/Recoverable: Minor penalty. 4. Penalize unstable or incoherent model reasoning. 5. Consider whether errors prevent goal completion.

Overall Trajectory Quality 1. Logical flow of steps, clarity of strategy, and coherence. 2. Balanced exploration vs. exploitation. 3. Closeness to final goal. 4. Reward consistent progress and coherent planning.

Output Format:
Return the evaluation as a JSON object: ``` { "index": [best_trajectory_index], "analysis": "Detailed reasoning explaining why this trajectory is the best, referencing progress, efficiency, loop detection, error severity, and overall quality." } ```

Input Prompt

Query: {query}

Trajectory 1: {trajectory_1}\n**Trajectory 2:** {trajectory_2}\n……\n**Trajectory N:** {trajectory_N}